Lesson plan · for the teacher

Unplugged, then plugged

Programming · Level 4 · Lesson 1

Time
40 min

Materials
Five large cards with 1, 2, 4, 8 and 16 dots; Chalk for a sorting network (optional); One phone with this lesson downloaded

Learning goals

- Choose unplugged activities that teach real ideas.
- Link each one to code soon after.
- Judge what unplugged can and can't do.

Activity: Binary cards

1. Warm-up (5 min): Ask how many numbers five fingers can show if each finger is up or down. Collect guesses; don't answer yet.
2. Play (15 min): Five pupils hold cards with 16, 8, 4, 2 and 1 dots. Call out numbers; the class tells them which cards to turn face up. Try 13, 31 and 0.
3. Name it (5 min): Explain that each card is a bit and that face up is 1: 13 is 01101. Five bits make 32 numbers, so the answer to the warm-up is 32.
4. Plug in (10 min): Show the same numbers on one phone in the Bits and bytes lab, then the Python snippet that adds the face-up cards.
5. Wrap-up (5 min): Ask what the cards could not show (letters, bigger numbers). Then give the exit ticket.

Make the cards from cardboard, large enough to see from the back row. One phone is enough for the plugged part.

Discussion

- What did pupils understand better because they acted it out?
- What can unplugged not teach?
- How soon after should the plugged lesson come?

Exit ticket

- Name one unplugged activity and the code you will link it to.

Try it in class tomorrow
Make binary cards (1, 2, 4, 8 and 16 dots) and stand five pupils at the front. Show numbers by turning cards over; then show the same numbers on one phone in the Bits and bytes lab.

Worksheet · for pupils

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Name _____ Class _____ Date _____

1 **Your class has just played binary cards. What is the best next step?**

◯ Play it again next week ◯ Link it to code on one phone ◯ Move to a new topic

2 **The halving method for binary, as Python: for 13 it prints 1101. Put the lines in order. One line doesn't belong.**

Number the lines in order (1, 2, 3…). Cross out any line that doesn't belong.

☐ `bits = str(n % 2) + bits`

☐ `n, bits = 13, ""`

☐ `n = n // 2`

☐ `n = n % 2`

☐ `while n > 0:`

☐ `print(bits)`

3 **The binary cards in Python. Which number do the face-up cards show?**

```
cards = [16, 8, 4, 2, 1]
up = [0, 1, 1, 0, 1]
total = 0
for i in range(5):
    total = total + cards[i] * up[i]
print(total)
```

◯ 13 ◯ 11 ◯ 5 ◯ 31

4 **Unplugged activities are enough on their own: pupils don't need to link them to code.**

◯ True ◯ False

5 **Match each unplugged activity to the computing idea it models.**

Binary cards	• • • •	Exact instructions in order: an algorithm
A sorting network	• • • •	Binary search
Guess my number by halving	• • • •	Comparing and swapping to sort
The human robot	• • • •	How numbers are stored in bits

Which unplugged activity could open your next computing unit? Write the activity, the idea it teaches, and how you will link it to code within a week.